



Second Semester

A.Y. 2025-2026

DISTURBANCES IN ENDOCRINE FUNCTIONS

In Partial Fulfillment of the Requirements in the Course
Care of Clients with Problems in Nutrition and Gastro-intestinal, Metabolism,
and Endocrine, Perception and Coordination (Acute and Chronic) - Lecture
BSN 32204

Submitted by: Group 4

Amac, Kurt Kaizzer
Begonia, Izabella Grace
Dela Rosa, Maria Cristina S.
Flaminiano, Alyssa Kirsten C.
Gabion, Terrence John F.
Manuel, Alexa Ysabel
Paralejas, Helaena V.
Sacro, Mikayella Marie A.
Torresyap, Ser Kyle V.

Submitted to:
Mr. Rainier Del Rosario, RN

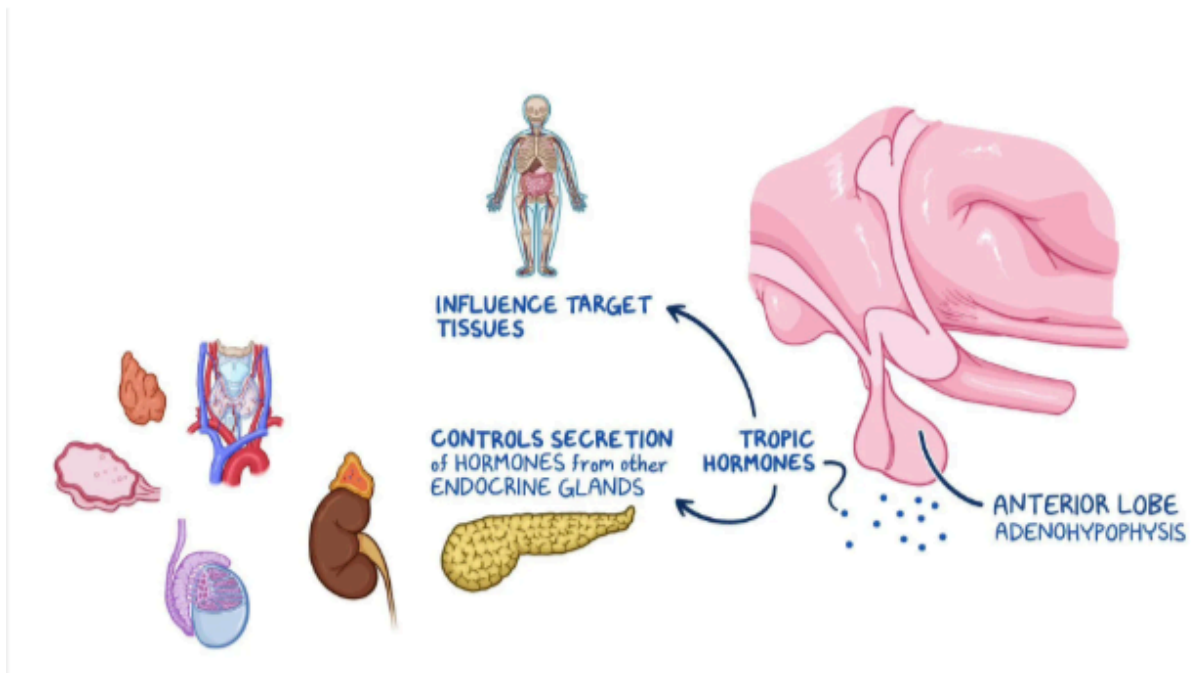


TABLE OF CONTENTS



INTRODUCTION

HYPERPITUITARISM



Hyperpituitarism is an endocrine disorder characterized by the excessive secretion of one or more hormones produced by the anterior pituitary gland. This condition most commonly results from pituitary adenomas, which are benign tumors that cause hormone overproduction and sometimes compress nearby brain structures. The effects of hyperpituitarism vary depending on the specific hormone involved but can significantly disrupt growth, metabolism, reproduction, and stress responses. If left untreated, it may lead to serious systemic complications affecting multiple organ systems.

Types:

Hyperpituitarism is classified according to the hormone that is excessively secreted:

1. Growth Hormone (GH) Hypersecretion: Causes **gigantism** in children (before epiphyseal closure) and acromegaly in adults.
2. Adrenocorticotrophic Hormone (ACTH) Hypersecretion: Leads to Cushing's disease, a form of **hypercortisolism**.
3. Prolactin Hypersecretion (Prolactinoma): Results in **hyperprolactinemia**, affecting reproductive function.
4. Thyroid-Stimulating Hormone (TSH) Hypersecretion: A rare cause of **secondary hyperthyroidism**.

Each type presents with distinct clinical features but shares a common origin in pituitary dysfunction.

Causes:

The most common cause of hyperpituitarism is a **benign pituitary adenoma**. These tumors may develop spontaneously or be associated with genetic conditions such as **Multiple Endocrine Neoplasia type 1 (MEN1)**. Other less common causes include **hypothalamic tumors**, ectopic hormone production, or pituitary hyperplasia. Hormonal overstimulation due to impaired feedback mechanisms can also contribute to excessive hormone release.

Clinical Manifestations:



Clinical manifestations depend on the hormone involved but may include both hormonal effects and mass effects of the tumor:

- GH excess: enlargement of hands and feet, coarse facial features, joint pain, cardiomegaly
- ACTH excess: truncal obesity, moon face, purple striae, muscle weakness, hypertension
- Prolactin excess: galactorrhea, amenorrhea, infertility, decreased libido
- TSH excess: weight loss, heat intolerance, tachycardia, tremors
- Mass effects: headaches, visual field defects (bitemporal hemianopia), nausea

Diagnosis:

Diagnosis is based on a combination of **laboratory testing and imaging studies**:

- **Serum hormone level assays** (e.g., GH, IGF-1, ACTH, prolactin)
- **Dynamic endocrine testing** such as glucose suppression tests for GH
- **MRI of the pituitary gland** to identify adenomas
- **Visual field testing** to assess optic chiasm compression

Early detection is essential to prevent irreversible complications.

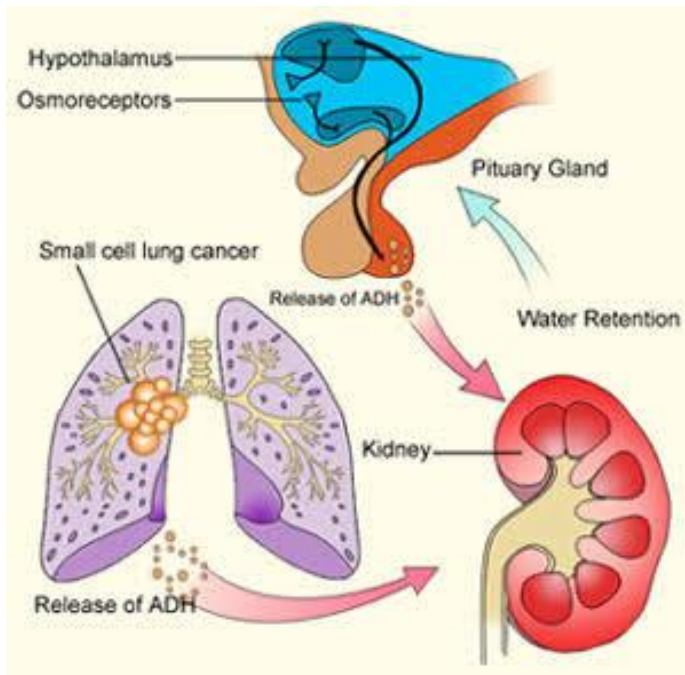
Nursing Managements:

Nursing care focuses on **monitoring, education, and collaborative management**:

- Perform thorough assessments of neurological status and endocrine symptoms
- Monitor vital signs, hormone levels, and intake/output
- Provide **pre- and post-operative care** for patients undergoing transsphenoidal surgery
- Educate patients about medication adherence, follow-up care, and symptom monitoring
- Offer emotional support and reinforce lifestyle modifications
- Coordinate care with endocrinologists, neurosurgeons, and other healthcare professionals

HYPOPITUITARISM

SYNDROME OF INAPPROPRIATE ANTIDIURETIC HORMONE SECRETION (SIADH)



Syndrome of inappropriate antidiuretic hormone secretion (SIADH) is a condition in which the body makes too much antidiuretic hormone (ADH). ADH is also called vasopressin. This hormone helps the kidneys regulate the amount of water your body loses through the urine. SIADH causes your body to retain too much water. This usually leads to a low blood sodium level (hyponatremia). SIADH is somewhat common. Hyponatremia, the main medical consequence of SIADH, is the most common electrolyte imbalance found in people in the hospital, and SIADH is the most common cause of the condition.

Causes:

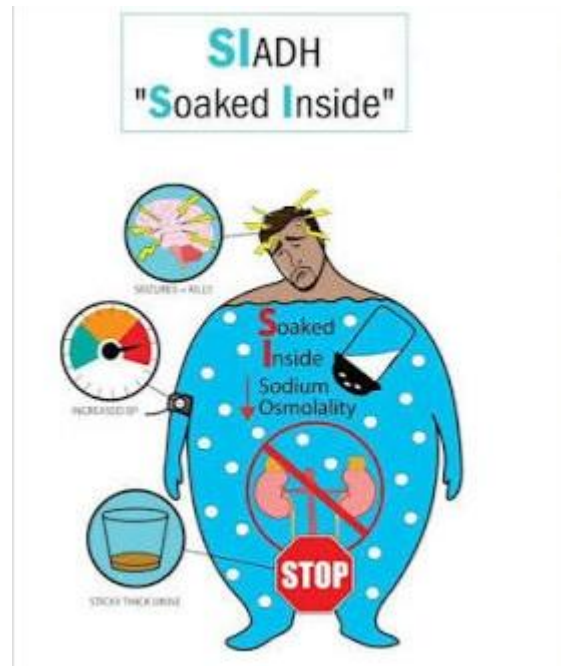
Conditions and situations that frequently lead to SIADH include:

- **Certain cancers:** Certain tumors can produce their own ADH, leading to excess ADH in your body and SIADH. Small cell lung cancer (SCLC) is the most common tumor leading to excess ADH production.
- **Certain medications:** Some drugs increase ADH release or mimic its effects on the kidneys. This causes the body to hold water inappropriately, diluting sodium in the blood, which results in SIADH. They include certain seizure drugs; antidepressants; cancer drugs; and heart, diabetes and blood pressure drugs.
- **Hormone deficiency:** Both hypopituitarism and hypothyroidism may lead to SIADH. Low levels of these certain hormones can disrupt the normal balance of ADH regulation. Without proper hormone signals, the body may release too much ADH, causing water retention and SIADH.

Clinical Manifestations:

In SIADH, urine is very concentrated because the body retains too much water. This extra water dilutes blood substances like sodium, causing hyponatremia, which is the main cause of symptoms. Mild low sodium may not cause problems, but moderate to severe hyponatremia can cause the following symptoms:

- Muscle cramps or weakness
- Nausea and vomiting





- Headache
- Problems with balance, which may result in falls.
- Mental changes such as confusion, memory problems, and/or strange behavior
- Seizure or coma (in severe cases)

Diagnosis

Laboratory Tests (blood and urine)

Blood

- Comprehensive Metabolic Panel (CMP): checks serum sodium, usually low in SIADH
- Serum Osmolality
- Other blood chemistry : Creatinine, bUN, Glucose, potassium, and uric acid help rule out other causes and assess kidney function

Urine

- Urine osmolality
- Urine Sodium

Functional Exclusion tests - because SIADH is a diagnosis of exclusion (other causes of hyponatremia must be ruled out)

- Thyroid function test (TSH): to rule out hypothyroidism
- Cortisol or ACTH stimulation tests : to exclude adrenal insufficiency

Clinical Assessment:

- Volume status exam : if the patient is euvolemic (normal fluid volume without dehydration or edema)
-

Prevention/Management

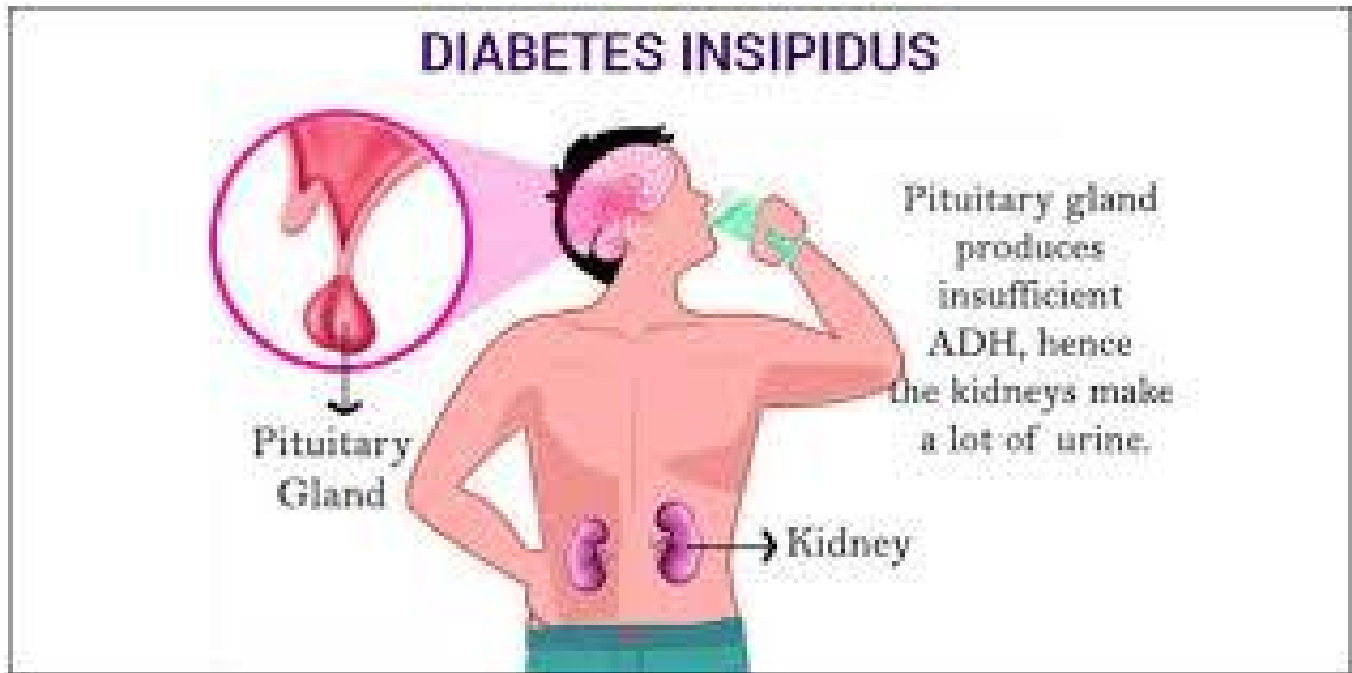
Nursing Management

- Implement fluid restriction - to help prevent further dilution of sodium in the blood and reduces the risk of worsening hyponatremia
- Ensure patient safety - since it can cause confusion and weakness, we should provide fall precautions, and apply seizure safety measure to prevent injury
- Provide patient education - educate patient about fluid restriction, dietary consideration, medication adherence, and warning signs of worsening hyponatremia
- Promote comfort and well-being - Encourage rest, reduce stress. To help patients cope with symptoms like fatigue, nausea, or mild confusion.

Pharmacological Management

- Vasopressin Receptor Antagonists - Blocks the effect of ADH on the kidneys, allowing excess water to be excreted and helping to raise serum sodium.
- Hypertonic Saline (3% NaCl) - Used through IV in severe symptomatic hyponatremia, such as confusion, seizures, or coma. It increases serum sodium.
- Demeclocycline - Antibiotic that reduces kidney responsiveness to ADH. It is used in chronic SIADH cases when fluid restriction fails.
- Loop Diuretics - Used in combination with sodium supplementation in certain cases to increase water excretion.

DIABETES INSIPIDUS



Definition

A disorder of the posterior pituitary gland characterized by deficiency of antidiuretic hormone (ADH) or Resistance of the kidneys to ADH. DI results in **excretion of large volumes of dilute urine, excessive thirst (polydipsia), fluid and electrolyte imbalance.**

Etiology / Causes:

1. Central (Neurogenic) DI = Due to decreased production or release of ADH

- **Causes:** Head trauma, Brain tumors (pituitary/hypothalamic), Neurosurgery, CNS infections (meningitis, encephalitis), Idiopathic, and Drugs affecting hypothalamus/pituitary

2. Nephrogenic DI = Due to kidney unresponsiveness to ADH

- **Causes:** Chronic kidney disease, Drugs (especially lithium), Hypercalcemia, Hypokalemia, Congenital disorders

3. Other Types

- **Gestational DI** – placental destruction of ADH
- **Primary polydipsia** – excessive fluid intake suppresses ADH



Pathophysiology:

Normal Function: ADH → increases water reabsorption in renal tubules

in DI: ↓ ADH OR kidney resistance → water not reabsorbed → massive urine output (up to 15–30 L/day) → increased plasma osmolality + dehydration

Clinical Manifestations:

- Urinary
- Polyuria (large volume, dilute urine)
- Urine specific gravity <1.005
- Nocturia
- Fluid Volume Deficit
- Extreme polydipsia (prefers cold water)
- Dry skin, dry mucous membranes
- Poor skin turgor
- Weight loss
- Hypotension
- Tachycardia
- Weak pulse
- Restlessness
- Irritability
- Confusion
- Seizures (severe cases)

Diagnostic Evaluation:

LABORATORY TESTS	DIAGNOSTIC TESTS	
• Urine specific gravity: ↓ (<1.005) • Urine osmolality: ↓ • Serum osmolality: ↑ • Serum sodium: ↑ (hypernatremia)	Water Deprivation Test: - Results: - Normal: urine concentrates - DI: remains dilute	
	Desmopressin (DDAVP) Test	
	Results	Interpretation
	Urine Concentrates	Central DI
	No Response	Nephrogenic DI



MEDICAL MANAGEMENT		
CENTRAL DI	NEPHROGENIC DI	FLUID THERAPY
Desmopressin (DDAVP) - drug of choice	Treat underlying cause	Oral fluids (if conscious)
Vasopressin (short-term use)	Thiazide Diuretics	IV fluids: - Hypotonic solutions (e.g., D5W) - Isotonic (if hypovolemic shock)
	Low-sodium diet	
	NSAIDs (reduce urine output)	
NURSING MANAGEMENT		
Assessment		
intake and output		
Monitor urine characteristics		
Daily weight		
Monitor serum sodium and osmolality Assess neurologic status.		

B. Nursing Management (HIGH-YIELD – SUDDARTH FOCUS)

Assessment

Strict intake and output (I&O)

Monitor urine characteristics

Daily weight

Monitor serum sodium & osmolality

Assess neurologic status

Interventions



Provide free access to water

Administer DDAVP as prescribed

Replace fluids as needed

Monitor for:

Dehydration

Hypovolemic shock

Safety

Monitor for water intoxication (overcorrection)

Seizure precautions (hyponatremia)

Patient Education

Importance of lifelong therapy (if chronic)

Maintain adequate fluid intake

Recognize:

Dehydration

Medication side effects

Complications

Severe dehydration

Hyponatremia

Hypovolemic shock

Acute kidney injury

HYPERTHYROIDISM



Hyperthyroidism, also called overactive thyroid, is a condition where your thyroid makes and releases high levels of thyroid hormone. It has multiple possible causes. The main thyroid hormones are triiodothyronine (T3), thyroxine (T4) and thyroid stimulating hormone (TSH).

Causes:

Graves' disease is the most common cause of hyperthyroidism. Your chances of developing it increase if you have a biological family history of thyroid disease and/or another autoimmune condition.

Consuming too much iodine (through foods or medications) can cause your thyroid to produce more thyroid hormone. Iodine is a mineral that your thyroid uses to create thyroid hormone. Receiving IV iodinated contrast (iodine “dye”) or taking amiodarone (a medication) may cause hyperthyroidism.

A growth in your pituitary gland that releases **excess thyroid-stimulating hormone (TSH)**. The pituitary gland normally releases TSH to trigger thyroid to release thyroid hormone. Excess TSH can lead to excess thyroid hormone. This is a very rare cause of hyperthyroidism.

Clinical Manifestations:

- Rapid heart rate (Tachycardia)
- Increased blood pressure
- Shakiness, like hand tremors
- Feeling anxious, nervous and/or irritable.
- Weight loss despite a regular or increased appetite.
- Diarrhea
- Increased sweating (diaphoresis) and sensitivity to warm temperatures.
- Hair loss or brittle hair.



- Difficulty sleeping (insomnia)
- Menstrual period changes
- Swelling and enlargement of the neck (goiter).
- Swelling or bulging of your eyes (thyroid eye disease)

Diagnosis:

Physical Examination

A physical exam is performed to identify visible and physical signs of thyroid dysfunction, such as an enlarged thyroid gland, rapid heart rate, and changes in skin temperature or moisture. These findings help the healthcare provider suspect hyperthyroidism early.

Thyroid Blood Tests

Thyroid blood tests are essential to confirm hyperthyroidism by measuring hormone levels. Elevated T3 and T4 levels, along with abnormal TSH levels, indicate excessive thyroid hormone production.

Imaging Tests

Imaging tests are used to determine the cause of hyperthyroidism and assess the structure and function of the thyroid gland. These tests help distinguish conditions such as Graves' disease, thyroid nodules, or inflammation.

Management:

Antithyroid medications such as methimazole and propylthiouracil (PTU) are used to reduce the production of thyroid hormones. They help control hyperthyroidism and relieve symptoms by bringing hormone levels back to normal without permanently damaging the thyroid.

Radioactive Iodine (RAI) Therapy is used to permanently treat hyperthyroidism by destroying overactive thyroid cells. This method effectively stops excessive hormone production and provides a long-term solution, although it usually results in hypothyroidism that requires lifelong hormone replacement.

Surgery (Thyroidectomy) is performed to quickly and definitively correct hyperthyroidism, especially when other treatments are not suitable. Removing part or all of the thyroid gland stops hormone overproduction but typically leads to hypothyroidism, requiring lifelong medication.

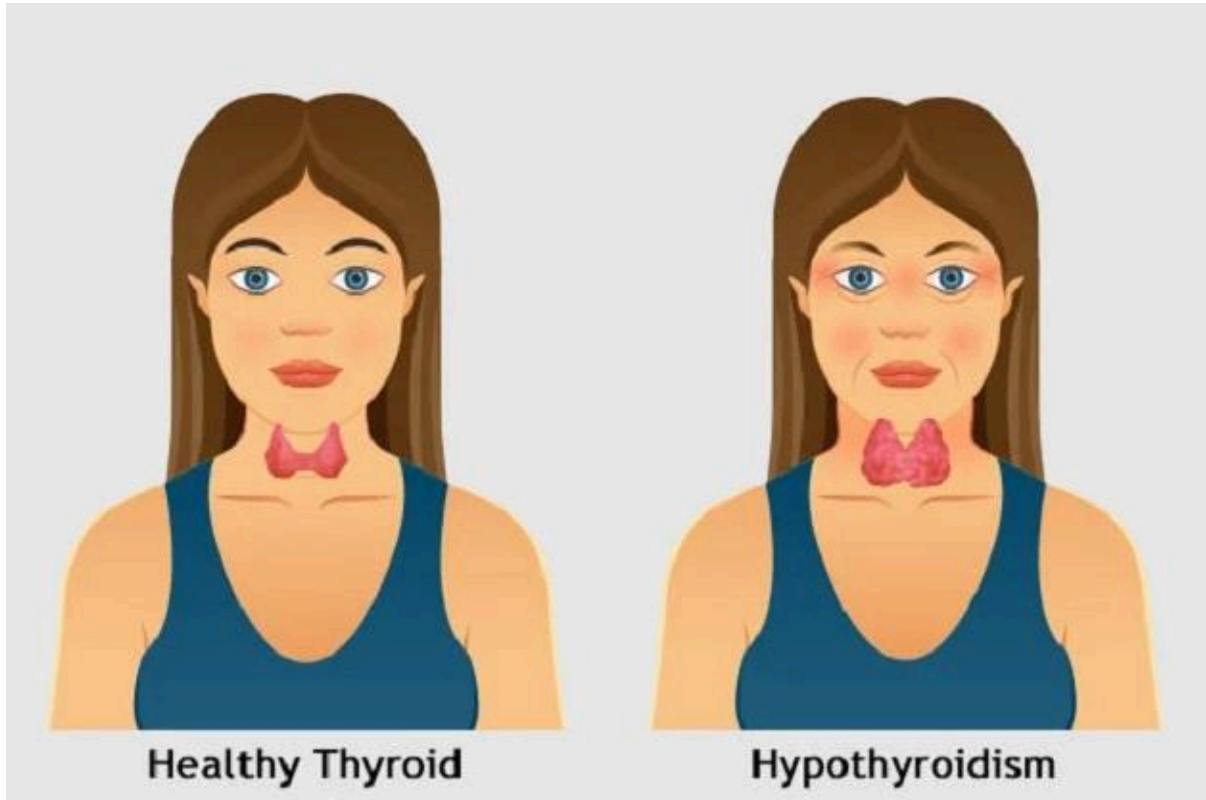
Beta-Blockers are used to control symptoms of hyperthyroidism, such as rapid heartbeat, tremors, and anxiety. They provide symptom relief but do not treat the underlying hormone imbalance, so they are used alongside other treatments.

Nursing Management:

- Promote rest – Space activities and provide frequent rest periods to prevent fatigue and excessive metabolic demand.
- Maintain a cool environment – Keep the room cool and well-ventilated to reduce heat intolerance and excessive sweating.
- Monitor vital signs – Regularly assess heart rate, blood pressure, and temperature to detect complications such as tachycardia or fever.
- Encourage adequate nutrition – Provide high-calorie, high-protein meals in small, frequent feedings to meet increased metabolic needs.
- Increase fluid intake – Encourage adequate fluid intake to prevent dehydration from excessive sweating and increased metabolism.

- Provide emotional support – Offer reassurance and a calm environment to reduce anxiety, irritability, and nervousness.
- Ensure eye protection – Protect the eyes from dryness and irritation, especially in patients with exophthalmos.

HYPOTHYROIDISM



Definition

Hypothyroidism is a condition caused by suboptimal levels of thyroid hormone, which can affect all body functions. Its severity ranges from mild or subclinical forms to myxedema, a life-threatening advanced form. The most common cause in adults is autoimmune thyroiditis (Hashimoto's disease), where the immune system attacks the thyroid gland.

Hypothyroidism can also develop in patients with a history of hyperthyroidism treated with radioiodine, antithyroid drugs, or thyroidectomy. Additionally, men who have received radiation therapy for head and neck cancer have an increased risk. Therefore, thyroid function testing is recommended in patients with these risk factors.

Causes

The thyroid is a tiny, butterfly-shaped gland situated under Adam's apple at the base of the neck. *Triiodothyronine (T-3)* and *thyroxine (T-4)* are the two primary hormones produced by the thyroid gland. Every cell in the body is impacted by these hormones. They facilitate the body's use of lipids and carbs. They aid in regulating body temperature. They have an impact on heart rate. Additionally, they aid in regulating the body's production of protein. When the thyroid gland produces insufficient amounts of hormones, hypothyroidism results.

Conditions or problems that can lead to Hypothyroidism are:

- Autoimmune disease
- Thyroid disease



- Radiation Therapy
- Thyroiditis
- Medicine
- Present at birth
- Pituitary disorder

Clinical Manifestations

- General / Metabolic
 - Extreme fatigue, decreased activity tolerance
 - Cold intolerance
 - Weight gain despite normal or decreased appetite
 - Subnormal body temperature (hypothermia in severe cases)
- Skin, Hair, and Nails
 - Dry, coarse, thickened skin (myxedema)
 - Hair thinning or hair loss
 - Brittle nails
 - Puffy, masklike facial appearance
- Neuromuscular
 - Muscle weakness
 - Numbness and tingling of fingers
 - Enlarged tongue
 - Increased size of hands and feet
 - Slow speech
- Cardiovascular
 - Bradycardia (slow pulse)
 - Decreased cardiac output and poor left ventricular function
 - Elevated serum cholesterol
 - Increased risk of atherosclerosis and coronary artery disease
 - Pericardial effusion (advanced cases)
- Gastrointestinal
 - Constipation
- Reproductive
 - More common in women (especially ages 40–70)
 - Menstrual disturbances (menorrhagia or amenorrhea)
 - Decreased libido
- Neuropsychiatric
 - Irritability early, progressing to apathy
 - Slowed mental processes
 - Cognitive impairment; dementia-like changes in advanced disease
 - Altered mental status in severe or untreated cases
- Respiratory
 - Hypoventilation
 - Sleep apnea
 - Respiratory muscle weakness
 - Pleural effusion (advanced cases)
- ENT / Sensory
 - Hoarseness or husky voice
 - Hearing impairment
- Medication and Surgical Sensitivity
 - Increased sensitivity to sedatives, opioids, and anesthetics
 - Increased surgical risk: hypotension, heart failure, altered mental status
- Population Considerations
 - More frequent in women and older adults



- Symptoms may be atypical or subtle in older adults

Assessment:

- **Physical examination.** The thyroid gland is inspected and palpated routinely in all patients. Assess thyroid gland for firmness (Hashimoto's) or tenderness (thyroiditis).
- **Thyroid antibodies.** Results of testing by immunoassay techniques for antithyroid antibodies are positive in Hashimoto's thyroiditis (100%)
- **TSH or Thyroid-stimulating hormone in the blood.** If the result is high, the test is done again, along with a blood test for thyroid hormone T4. If the result appear that TSH is high and T4 is low, then it signifies hypothyroidism.

Diagnosis:

- Activity intolerance related to fatigue and depressed cognitive process.
- Risk for imbalanced body temperature related to cold intolerance.
- Constipation related to depressed gastrointestinal function.
- Ineffective breathing pattern related to depressed ventilation.
- Disturbed thought processes related to depressed metabolism and altered cardiovascular and respiratory status.

Management:

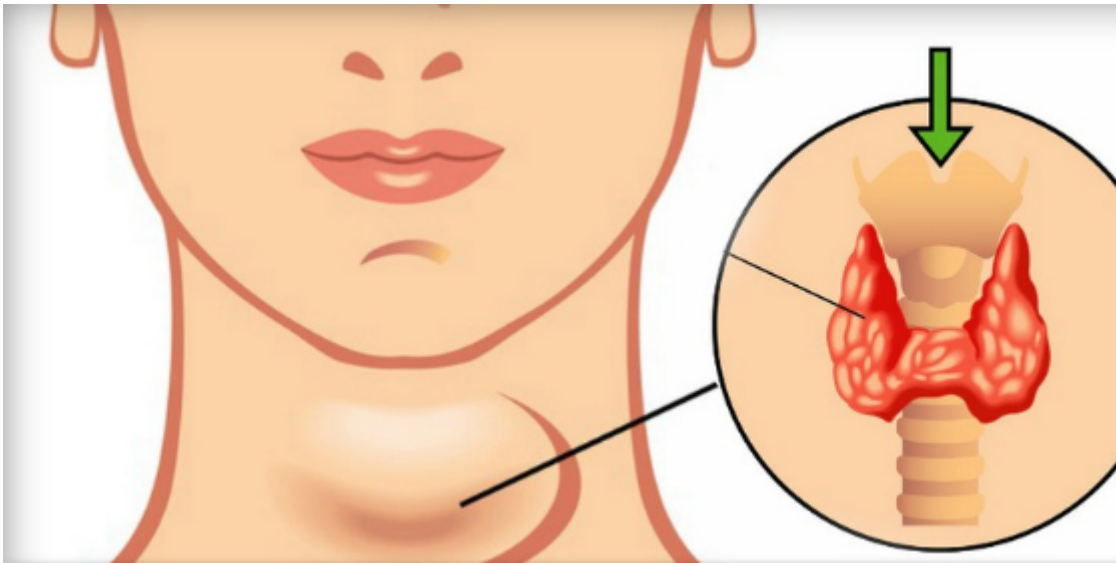
Pharmacologic Therapy

- *Synthetic levothyroxine* (Synthroid or Levothroid) is commonly prescribed for treating hypothyroidism and suppressing nontoxic goiters. The dosage is based on the patient's serum TSH concentration. If replacement therapy is adequate, the symptoms of myxedema disappear and normal metabolic activity is resumed.
- If the disease progresses to myxedema coma, IV administration of T4 and T3 are advised rather than T4 alone. T4 is administered IV until the patient is able to safely take T4 in oral form. T3 is administered IV until the patient is stable (Ross, 2016a).
- Administration of high-dose glucocorticoids (hydrocortisone) every 8 to 12 hours for 24 hours followed by low-dose therapy is recommended until coexisting adrenal insufficiency is ruled out (Ross, 2016a). Serum levels of T4, TSH, and cortisol should be obtained before and after

Nursing Intervention

- Promote rest - Space activities to promote rest and exercise as tolerated.
- Protect against coldness - Provide an extra layer of clothing or extra blanket.
- Avoid external heat exposure - Discourage and avoid the use of external heat sources.
- Mind the temperature - Monitor patient's body temperature.
- Increase fluid intake - Encourage increased fluid intake within the limits of fluid restriction.
- Provide foods high in fiber.

HYPERPARATHYROIDISM



Hyperparathyroidism is a disorder in which the **parathyroid glands produce excessive amounts of parathyroid hormone (PTH)**, leading to **hypercalcemia** and disturbances in calcium and phosphorus balance. Elevated PTH causes increased calcium release from bones, increased intestinal absorption, and decreased renal excretion of calcium. Over time, this condition can result in **bone demineralization, kidney damage, and neuromuscular dysfunction**.

Types:

Hyperparathyroidism is classified into three major types:

1. **Primary Hyperparathyroidism:** Caused by intrinsic abnormalities of the parathyroid glands, usually an adenoma.
2. **Secondary Hyperparathyroidism:** Occurs as a compensatory response to chronic hypocalcemia, commonly due to chronic kidney disease or vitamin D deficiency.
3. **Tertiary Hyperparathyroidism:** Results from prolonged secondary hyperparathyroidism leading to autonomous PTH secretion.

Causes:

The most common cause of primary hyperparathyroidism is a **parathyroid adenoma**, followed by gland hyperplasia. Secondary hyperparathyroidism is often caused by **chronic renal failure**, malabsorption syndromes, or inadequate vitamin D levels. Rarely, parathyroid carcinoma may be responsible. Long-term calcium imbalance disrupts normal bone and renal function.

Clinical Manifestations:

Symptoms range from mild to severe and are often summarized as “**stones, bones, groans, and psychiatric overtones**”:

- **Renal:** kidney stones, polyuria, dehydration
- **Skeletal:** bone pain, fractures, osteoporosis
- **Gastrointestinal:** constipation, nausea, abdominal pain
- **Neuromuscular:** muscle weakness, fatigue
- **Neuropsychiatric:** depression, confusion, memory problems

Some patients may be asymptomatic and diagnosed incidentally.

Diagnosis:

Diagnosis relies on biochemical and imaging studies:

- **Elevated serum calcium levels**
- **Increased PTH levels**
- **Low or normal phosphate levels**
- **24-hour urine calcium test**
- **Bone density testing (DEXA scan)**
- **Imaging studies** (ultrasound or sestamibi scan) for surgical planning

Nursing Management:

Nursing management emphasizes **monitoring, patient education, and prevention of complications**:

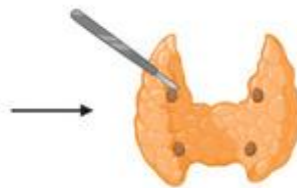
- Monitor calcium levels and observe for signs of hypercalcemia
- Encourage adequate hydration to reduce kidney stone formation
- Implement fall prevention strategies due to bone fragility
- Provide **pre- and post-operative care** for parathyroidectomy
- Educate patients about dietary calcium, vitamin D intake, and medication compliance
- Collaborate with the healthcare team for long-term management

HYPOPARATHYROIDISM

Normoparathyroidism



Normocalcaemia
Normophosphataemia
Normal PTH levels



SURGICAL RESECTION

Hypoparathyroidism



Hypocalcaemia
Hyperphosphataemia
Low/absent PTH levels

DEFINITION:

- **Hypoparathyroidism** is a rare endocrine disorder characterized by insufficient production of parathyroid hormone (PTH) by the parathyroid glands. This deficiency results in low calcium levels in the blood (hypocalcemia) and high phosphorus levels (hyperphosphatemia). The imbalance can affect the nervous system, muscles, bones, and teeth because calcium is essential for nerve conduction, muscle contraction, and bone strength.

CAUSES:

Common causes of hypoparathyroidism include:



1. **Damage to your parathyroid glands** – About 75% of cases are caused by accidental damage to the parathyroid glands during neck or thyroid surgery. This usually happens soon after surgery, but it can also occur many years later. In rare cases, radiation therapy can damage your parathyroid glands.
2. **Certain autoimmune disorders** – Certain diseases cause your immune system to attack your parathyroid glands.
3. **Certain genetic disorders** – Less than 10% of cases are genetic. The most common is DiGeorge syndrome, where babies are born without parathyroid glands, leading to lifelong low calcium. About 60% of children with hypoparathyroidism have DiGeorge syndrome.
4. **Low levels of magnesium**– Low magnesium can cause the parathyroid glands to stop making or releasing enough PTH, leading to low calcium levels.
5. **Infiltrative diseases** – conditions like iron overload or cancer affecting the glands.

CLINICAL MANIFESTATIONS

The signs and symptoms of hypoparathyroidism include:

- Tingling or numbness in your lips, fingers and toes
- Muscle cramps and spasms
- “Brain fog” or confusion
- Abnormal heart rhythm
- Brittle nails and dry skin
- Cataracts
- Weak tooth enamel (in children)

In most cases, hypoparathyroidism progresses gradually, and symptoms can be mild. Many people have symptoms for years before they get a diagnosis.

DIAGNOSIS

Hypoparathyroidism is generally diagnosed when a person has low levels of calcium and parathyroid hormone in their blood. Healthcare providers sometimes “accidentally” find hypoparathyroidism when a routine blood screening shows you have low levels of blood calcium.

If you’re experiencing symptoms of hypoparathyroidism, your healthcare provider will perform a physical exam and ask questions about your symptoms and medical history.

They may order one or more of the following tests to help diagnose hypoparathyroidism:

- Parathyroid hormone (PTH) blood test
- Calcium blood test
- Phosphorus blood test
- Magnesium blood test
- 24-hour urine calcium test

Other tests conducted to check for more serious side effects of hypoparathyroidism includes:

- An electrocardiogram (ECG or EKG) to check for an abnormal heart rhythm
- A computed tomography (CT) scan to check for calcium deposits in your brain

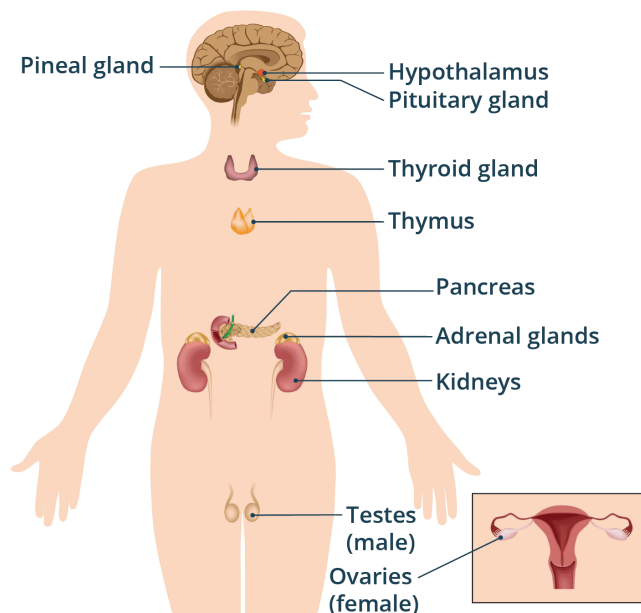
MANAGEMENT/PREVENTION

The goal of treatment for hypoparathyroidism is to get the right amount of calcium and minerals in your body.

Treatment can include:

- **Taking calcium carbonate and vitamin D supplements:** You have to take calcium and vitamin D supplements for life. Your body needs vitamin D to absorb and use calcium, so you need to take both supplements. Some people also need a magnesium supplement.
- **Eating a high-calcium and low-phosphorus diet:** Your healthcare provider may recommend eating less red meat and eating more calcium-rich foods like broccoli, yogurt and tofu. They may also ask you to limit processed foods, which usually contain phosphorus. A dietitian may give you specific instructions on what foods to avoid.
- **Getting parathyroid hormone (PTH) injections:** If taking calcium and vitamin D supplements isn't working to treat hypoparathyroidism, your provider might have you take parathyroid hormone injections.
- **Thiazide diuretics:** This medication can help reduce how much calcium you lose in your pee.

CUSHING'S DISEASE



DEFINITION:

Cushing's syndrome is an endocrine disorder caused by prolonged exposure to abnormally high levels of cortisol, a hormone produced by the adrenal glands that is essential for regulating metabolism, immune function, blood pressure, and the body's response to stress. The condition may result from excessive cortisol production within the body due to disorders of the hypothalamic-pituitary-adrenal axis, such as pituitary or adrenal tumors, or from long-term use of corticosteroid medications. Chronic cortisol excess leads to significant metabolic, physical, and psychological changes, including characteristic fat



redistribution, muscle weakness, immune suppression, and increased risk for complications such as hypertension, diabetes, osteoporosis, and cardiovascular disease. Early identification and appropriate management are crucial to prevent long-term morbidity and improve patient outcomes.

CAUSES:

1. Exogenous (Most Common)

- Long-term use of **glucocorticoid medications** (e.g., prednisone, dexamethasone) for conditions like asthma, autoimmune diseases, or transplant therapy.

2. Endogenous

- **Pituitary adenoma** producing excess ACTH → stimulates adrenal glands (Cushing disease).
- **Ectopic ACTH production** by non-pituitary tumors (e.g., small-cell lung cancer).
- **Adrenal tumors** (adenomas or carcinomas) producing cortisol independently.
- **Adrenal hyperplasia** or rare genetic/familial endocrine disorders.

CLINICAL MANIFESTATIONS:

Common Signs & Symptoms

- Weight gain (especially trunk and face)
- Moon face, buffalo hump
- Purple/pink stretch marks (striae) on abdomen, hips, thighs
- Thin skin, easy bruising
- Slow wound healing
- Muscle weakness, fatigue
- Acne

Additional Manifestations

- Women: hirsutism (excess facial/body hair), irregular periods
- Men: lower libido, fertility issues
- Neuropsychiatric: depression, irritability, poor concentration
- Cardiometabolic: hypertension, hyperglycemia/diabetes
- Increased infections, osteoporosis, bone fractures

DIAGNOSIS:

1. Screening Tests

- **24-hour urinary free cortisol** – elevated in Cushing's.
- **Late-night salivary cortisol** – loss of normal cortisol rhythm.
- **Low-dose dexamethasone suppression test** – failure to suppress cortisol indicates Cushing's.

2. Determining Cause

- **ACTH level measurement** – high levels suggest ACTH-dependent causes (pituitary/ectopic).
- **High-dose dexamethasone suppression test or CRH stimulation test** – helps differentiate pituitary vs ectopic sources.
- **Imaging** (MRI/CT) to locate pituitary or adrenal tumors.

MANAGEMENT/PREVENTION:

A. Treat Underlying Cause

- Surgery
- Pituitary: transsphenoidal resection of adenoma.
- Adrenal tumor removal.
- Resection of ectopic source when possible.
- Radiation therapy – for residual pituitary disease.

B. Medical Therapies

- Medications to reduce cortisol production or block its effects (used when surgery isn't possible or incomplete)

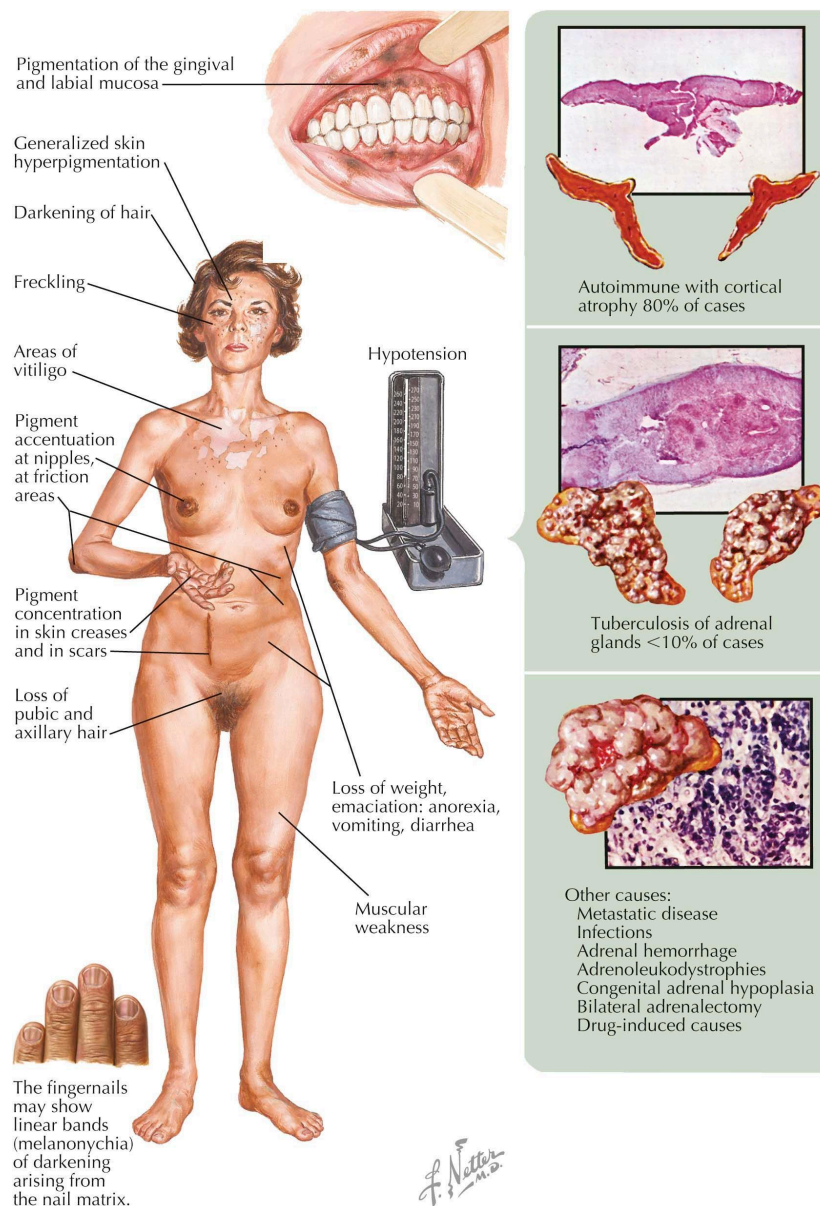
C. Supportive & Symptomatic Care

- Control hypertension, diabetes, osteoporosis
- Treat infections, psychological support
- Lifelong follow-up due to risk of recurrence or adrenal insufficiency post-treatment

Prevention

- Avoid unnecessary long-term glucocorticoid use or use lowest effective dose.
- Regular monitoring for symptoms in patients on steroids.
- Educate about signs of recurrence after treatment.

ADDISON'S DISEASE





Addison's disease, also known as chronic primary adrenocortical insufficiency, is a disorder resulting from progressive destruction or dysfunction of the adrenal cortex, leading to deficient production of glucocorticoids (cortisol), mineralocorticoids (aldosterone), and adrenal androgens.

The adrenal glands possess a substantial functional reserve, and therefore clinical symptoms typically do not manifest until approximately 90% or more of bilateral adrenal cortical function has been lost. The resulting hormonal deficiencies disrupt metabolism, electrolyte balance, vascular tone, immune response, and stress adaptation, producing characteristic systemic and cutaneous findings.

Causes:

Addison's disease is caused by primary adrenal gland failure, meaning the pathology originates within the adrenal glands themselves.

1. Autoimmune Causes

Autoimmune adrenalitis is the most common cause of Addison's disease in developed countries. In this condition, the body's immune system mistakenly recognizes components of the adrenal cortex as foreign and mounts an immune response against them. This results in progressive, immune-mediated destruction of adrenal cortical tissue, ultimately leading to reduced production of cortisol, aldosterone, and adrenal androgens.

Autoimmune adrenalitis frequently occurs as part of autoimmune polyglandular syndromes, in which multiple endocrine organs are affected by autoimmune processes. The presence of these associated conditions increases the likelihood of autoimmune adrenal insufficiency. Common associated conditions include:

- **Type 1 diabetes mellitus**, due to autoimmune destruction of pancreatic beta cells
- **Autoimmune thyroid disease** (such as Hashimoto thyroiditis or Graves disease)
- **Pernicious anemia**, caused by autoimmune destruction of gastric parietal cells leading to vitamin B12 deficiency

2. Infectious Causes

Infectious diseases can cause Addison's disease through direct invasion and destruction of adrenal tissue.

- **Tuberculosis** is one of the most common infectious causes worldwide, particularly in endemic regions. *Mycobacterium tuberculosis* progressively infiltrates and destroys both adrenal glands. This leads to chronic primary adrenal insufficiency.
- Other infectious causes include:
 - **Fungal infections** (e.g., histoplasmosis) that invade adrenal tissue.
 - **Opportunistic infections** in immunocompromised individuals, including those with HIV/AIDS.
- In contrast to chronic infections, acute adrenal failure may occur during severe bacterial infections. **Meningococcal septicemia** is a classic cause which leads to bilateral adrenal hemorrhage, known as **Waterhouse - Friderichsen syndrome**. This results in sudden, severe, and life-threatening adrenal insufficiency.

3. Other Causes

Several non-autoimmune, non-infectious conditions may also result in Addison's disease by damaging or removing functional adrenal tissue:

- **Metastatic malignancy** may infiltrate both adrenal glands, most commonly from primary cancers of the lung, breast, or gastrointestinal tract, leading to destruction of adrenal tissue.
- **Bilateral adrenal hemorrhage or infarction** may occur due to trauma, anticoagulant therapy, sepsis, or severe hypotension, resulting in sudden loss of adrenal function.



- **Surgical removal of both adrenal glands** (bilateral adrenalectomy), performed for certain adrenal or systemic diseases, causes permanent adrenal insufficiency.
- **Medications that inhibit adrenal steroid synthesis**, such as ketoconazole or mitotane, interfere with cortisol and aldosterone production and may lead to adrenal insufficiency when used long-term or at high doses.

Clinical Manifestations

1. General and Systemic Manifestations

- Affects males and females equally.
- Lethargy and generalized malaise are often the earliest symptoms.
- Symptoms may remain subtle until the patient experiences a major stressor (e.g., infection, trauma, surgery).
- Fatigue and muscle weakness can be severe, sometimes making speaking difficult.
- Unintentional weight loss and anorexia are common.
- Dehydration occurs due to sodium and water loss.
- Hypotension, often orthostatic, is frequently present.
- Small cardiac silhouette may be observed on chest radiography due to decreased blood volume.
- Neuropsychiatric symptoms may include excessive nervousness, emotional lability, depression, and decreased concentration.

2. Cutaneous and Pigmentary Manifestations

- Generalized hyperpigmentation ("bronze" discoloration) is characteristic of primary adrenal insufficiency.
- Pigmentation is accentuated in areas of friction (waistline, elbows, knees), groin, nipples, and scrotum.
- Palmar and plantar creases appear darkened.
- Hyperpigmentation of scars is often present.
- Pigmentary changes of gingival and labial mucosa may occur.
- Vitiligo may coexist, especially in autoimmune cases.
- Increased hair pigmentation is possible, usually subtle and slowly progressive.
- Pigmentary changes are absent in secondary adrenal insufficiency caused by pituitary dysfunction.

3. Hair and Sexual Characteristics

- Marked loss of axillary and pubic hair occurs due to adrenal androgen deficiency.
- Hair loss is more pronounced in females.
- Males retain partial androgen production from the testes, so hair loss is less severe.

Diagnosis

1. Laboratory Findings

- Low serum cortisol levels.
- Hyponatremia due to sodium loss.
- Hyperkalemia due to impaired potassium excretion.
- Evidence of volume depletion from aldosterone deficiency.

2. Diagnostic Testing

- **ACTH (cosyntropin) stimulation test** confirms the diagnosis.
- Synthetic ACTH is administered intravenously.
- Serum cortisol is measured before and after administration.
- In Addison's disease, serum cortisol fails to increase appropriately.

Treatment

1. Hormone Replacement Therapy

- Hydrocortisone is used to replace deficient cortisol.
- Fludrocortisone is used to replace aldosterone and maintain blood pressure and electrolyte balance.



2. Additional Management

- Treat underlying causes, such as infections or autoimmune disease.
- Increase glucocorticoid dosing during illness, surgery, or trauma.
- Educate patients about adrenal crisis and the importance of lifelong therapy.

Prevention

- Early recognition and treatment of autoimmune adrenalitis.
- Prompt diagnosis and treatment of infectious causes, particularly tuberculosis.
- Careful monitoring of patients with known autoimmune endocrine disorders.
- Educate patients about stress-dose steroid requirements and signs of adrenal crisis.
- Lifelong follow-up to ensure adequate hormone replacement and prevent complications.

DIABETES MELLITUS

Diabetes is a condition that happens when the blood sugar (glucose) level is too high. It develops when the pancreas doesn't make enough insulin or any at all, or when the body isn't responding to the effects of insulin properly. Diabetes affects people of all ages. Most forms of diabetes are chronic (lifelong), and all forms are manageable with medications and/or lifestyle changes.

Causes:

Insulin Resistance

Type 2 diabetes mainly results from insulin resistance. Insulin resistance occurs when cells in the muscles, fat, and liver don't respond properly to insulin. Several factors contribute to insulin resistance, including obesity, lack of physical activity, poor diet, hormonal imbalances, genetics, and certain medications.

Autoimmune Disease

Type 1 diabetes and LADA develop when the immune system attacks the insulin-producing cells in the pancreas.

Hormonal Imbalances

During pregnancy, the placenta releases hormones that cause insulin resistance. Gestational diabetes may develop if the pancreas cannot produce enough insulin to overcome this resistance. Other hormone-related conditions, such as acromegaly and Cushing syndrome, can also lead to Type 2 diabetes.

Pancreatic Damage

Physical damage to the pancreas—caused by disease, surgery, or injury—can impair insulin production, resulting in Type 3c diabetes.

Genetic Mutations

Certain genetic mutations can cause maturity-onset diabetes of the young (MODY) and neonatal diabetes.

Types:

Type 2 Diabetes

With this type, the body doesn't make enough insulin and/or the body's cells don't respond normally to insulin (insulin resistance). This is the **most common type of diabetes**. It mainly affects adults, but children can develop it as well.



Prediabetes

This type is the stage before Type 2 diabetes. The blood glucose levels are higher than normal but not high enough to be officially diagnosed as Type 2 diabetes.

Type 1 Diabetes

This type is an autoimmune disease in which the immune system attacks and destroys insulin-producing cells in the pancreas for unknown reasons. Up to 10% of people with diabetes have Type 1 diabetes. It is usually diagnosed in children and young adults, but it can develop at any age.

Gestational Diabetes

This type develops in some people during pregnancy. Gestational diabetes usually goes away after pregnancy. However, individuals who experience gestational diabetes have a higher risk of developing Type 2 diabetes later in life.

Type 3c Diabetes

This form of diabetes occurs when the pancreas experiences damage (other than autoimmune damage), affecting its ability to produce insulin. Conditions such as pancreatitis, pancreatic cancer, cystic fibrosis, and hemochromatosis can lead to pancreatic damage that causes diabetes. Removal of the pancreas (pancreatectomy) also results in Type 3c diabetes.

Latent Autoimmune Diabetes in Adults (LADA)

Similar to Type 1 diabetes, LADA results from an autoimmune reaction, but it develops much more slowly. People diagnosed with LADA are usually over the age of 30.

Maturity-Onset Diabetes of the Young (MODY)

MODY, also known as monogenic diabetes, occurs due to an inherited genetic mutation that affects how the body makes and uses insulin. There are more than 10 identified types of MODY. It affects up to 5% of people with diabetes and commonly runs in families.

Neonatal Diabetes

This is a rare form of diabetes that occurs within the first six months of life and is also a type of monogenic diabetes. About 50% of affected infants have a lifelong form called permanent neonatal diabetes mellitus. In the remaining cases, the condition resolves within a few months after onset but may recur later in life. This is known as transient neonatal diabetes mellitus.

Brittle Diabetes

Brittle diabetes is a severe form of Type 1 diabetes characterized by frequent and extreme episodes of high and low blood sugar levels. This instability often leads to repeated hospitalizations. In rare cases, a pancreas transplant may be required for permanent treatment.

Clinical Manifestations:

- Increased thirst (polydipsia) and dry mouth.
- Frequent urination.
- Fatigue.
- Blurred vision.
- Unexplained weight loss.
- Numbness or tingling in your hands or feet.
- Slow-healing sores or cuts.



- Frequent skin and/or vaginal yeast infections.

Diagnosis:

Fasting Blood Glucose Test

For this test, the patient does not eat or drink anything except water (fast) for at least eight hours before the test. Because food can greatly affect blood sugar, this test shows the baseline blood glucose level.

Random Blood Glucose Test

“Random” means the test can be done at any time, regardless of fasting.

A1c Test (HbA1c or Glycated Hemoglobin Test)

This test provides the average blood glucose level over the past two to three months.

Oral Glucose Tolerance Test (OGTT)

To screen for and diagnose gestational diabetes, providers order this test.

Blood Glucose Chart

Mg/DL	Fasting	After Eating	2 - 3 Hours After Eating
Normal	80 - 100	170 - 200	120 - 140
Impaired Glucose	101 - 125	190 - 230	140 - 160
Diabetic	126+	220 - 300	200+

Management:

Blood Sugar Monitoring

Monitoring the blood sugar (glucose) is key to determining how well the current treatment plan is working. It provides information on how to manage diabetes on a daily—or sometimes hourly—basis. Blood sugar can be monitored through frequent checks with a glucose meter and finger stick and/or with a continuous glucose monitor (CGM). Healthcare providers determine the best blood sugar range for the patient.

Oral Diabetes Medications

Oral diabetes medications (taken by mouth) help manage blood sugar levels in individuals who still produce some insulin—mainly people with Type 2 diabetes and prediabetes. People with gestational diabetes may also require oral medications. There are several types, with metformin being the most common.

Insulin

People with Type 1 diabetes require synthetic insulin injections to live and manage their condition. Some people with Type 2 diabetes also need insulin. Different types of insulin vary in how quickly they start working and how long they last. Insulin can be administered via syringe (shot), insulin pens, insulin pumps, or rapid-acting inhaled insulin.

Diet

Meal planning and following a healthy diet are essential for diabetes management, as food significantly affects blood sugar. For those taking insulin, carbohydrate counting is crucial because the amount of carbs consumed determines the required insulin dose. Healthy eating habits also support weight management and reduce the risk of heart disease.



Exercise

Physical activity increases insulin sensitivity and helps reduce insulin resistance. Regular exercise is an important part of diabetes management for all individuals with diabetes.

Nursing Management:

- Promote blood sugar monitoring – Assist the patient in checking blood glucose regularly and recording results to guide treatment.
Encourage adherence to medications – Ensure the patient takes oral diabetes medications or insulin as prescribed and understands proper administration.
- Promote healthy diet – Provide meals according to diabetic dietary recommendations and educate on carbohydrate counting if using insulin.
- Encourage regular exercise – Plan physical activity as tolerated to improve insulin sensitivity and overall health.
Prevent hypoglycemia and hyperglycemia – Monitor for signs of low or high blood sugar and intervene promptly when necessary.
- Provide emotional support – Offer counseling and reassurance to reduce anxiety and stress related to diabetes management.
- Promote foot care – Educate and assist with daily inspection and care of feet to prevent ulcers and infections.

REFERENCES:

- StatPearls. (2024). *Hyperpituitarism*. In StatPearls [Internet]. StatPearls Publishing.
- Cleveland Clinic. (2024). *Hyperpituitarism (overactive pituitary gland)*. Cleveland Clinic Health Library.
- Cleveland Clinic. (2024, October 31). Hyperthyroidism. Cleveland Clinic. <https://my.clevelandclinic.org/health/diseases/14129-hyperthyroidism>
- Cleveland Clinic. (2024). *Hypoparathyroidism: Symptoms and causes*. <https://my.clevelandclinic.org/health/diseases/22672-hypoparathyroidism>
- Mayo Clinic Staff. (2022, December 10). Hypothyroidism (underactive thyroid): Symptoms and causes. Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/hypothyroidism/symptoms-causes/syc-2035028>
- Ngocanh. (2024, January 9). Hypothyroidism: Insights into causes, recognition and treatment. Benhvien Thu Cuc. Retrieved September 19, 2025, from <https://en.benhvienthucuc.vn/hypothyroidism-insights-into-causes-recognition-and-treatment/>
- Mayo Clinic Staff. (2025). *Hyperparathyroidism: Symptoms and causes*. Mayo Clinic.
- StatPearls. (2025). *Primary hyperparathyroidism*. In **StatPearls [Internet]**. StatPearls Publishing.
- Anderson, B. E. (2023). *The Netter Collection of Medical Illustrations: Integumentary System* (Vol. 4). Elsevier Health Sciences.
- Kairys, N., Anastasopoulou, C., & Schwell, A. (2023, February 27). Cushing disease. StatPearls - NCBI Bookshelf. <https://www.ncbi.nlm.nih.gov/books/NBK448184/>



- Cushing syndrome - Symptoms and causes. (n.d.). Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/cushing-syndrome/symptoms-causes/syc-20351310>
- Johns Hopkins Medicine. (n.d.). Hypoparathyroidism. <https://www.hopkinsmedicine.org/health/conditions-and-diseases/hypoparathyroidism>
- Mayo Clinic Staff. (2024, December 21). *Addison's disease – Symptoms and causes*. Mayo Clinic. <https://www.mayoclinic.org/diseases-conditions/addisons-disease/symptoms-causes/syc-20350293>
- National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK). (2024). *Definition & facts of adrenal insufficiency & Addison's disease*. <https://www.niddk.nih.gov/health-information/endocrine-diseases/adrenal-insufficiency-addisons-disease/definition-facts>
- U.S. National Library of Medicine. (n.d.). *Addison disease*. In *StatPearls*. <https://www.ncbi.nlm.nih.gov/books/NBK441994/>